

Problem 3.4 A 300-N force is applied at A as shown. Determine (a) the moment of the 300-N force about D, (b) the smallest force applied at B that creates the same moment about D.

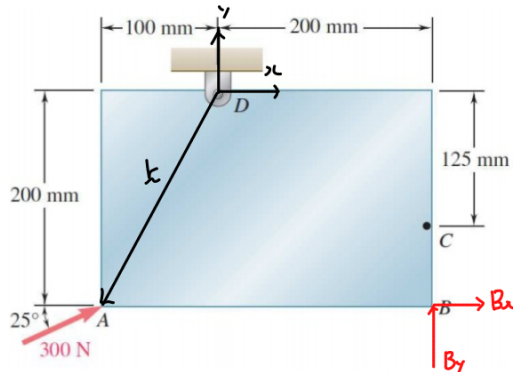


Fig. P3.4

$$\underline{r}_D = (-100\text{ mm})\hat{i} + (-200\text{ mm})\hat{j}$$

$$\underline{F}_A = (300\cos 25^\circ)\hat{i} + (300\sin 25^\circ)\hat{j}$$

$$= (271.89\text{ N})\hat{i} + (126.79\text{ N})\hat{j}$$

$$\underline{M}_D = \underline{r}_D \times \underline{F}_A$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -100\text{ mm} & -200\text{ mm} & 0 \\ 271.89\text{ N} & 126.79\text{ N} & 0 \end{vmatrix}$$

$$a) = (41.7\text{ N}\cdot\text{m})\hat{k}$$

$$\underline{r}_B = (200\text{ mm})\hat{i} + (-200\text{ mm})\hat{j}$$

$$202.0809 \approx 202$$

$$\underline{F}_B = (B_x\text{ N})\hat{i} + (B_y\text{ N})\hat{j}$$

$$\underline{M}_D = \underline{r}_B \times \underline{F}_B$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 200\text{ mm} & -200\text{ mm} & 0 \\ B_x\text{ N} & B_y\text{ N} & 0 \end{vmatrix} = (41.7\text{ N}\cdot\text{m})\hat{k}$$

$$(0.2B_y + 0.2B_x)\hat{k} = 41.7\hat{k}$$

$$B_x + B_y = 208.5$$

$$0.2B_y + 0.2B_x = 41.7$$

$$F_B = \sqrt{B_x^2 + B_y^2} = \sqrt{2B_x^2 - 417B_x + (208.5)^2} \text{ in N}$$

$$\rightarrow B_x = \frac{417}{4}\text{ N} \quad F_B = 147.4\text{ N} \angle 45^\circ$$

$$B_y = \frac{417}{4}\text{ N}$$

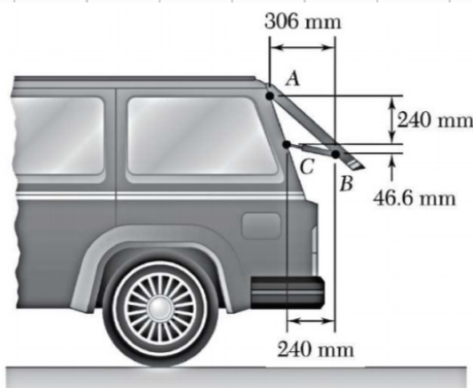
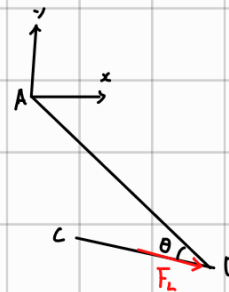


Fig. P3.11

Problem 3.11 The tailgate of a car is supported by the hydraulic lift BC. If the lift exerts a 500-N force directed along its centerline on the ball and socket at B, determine the moment of the force about A.



$$\overline{AB} = \sqrt{(306\text{ mm})^2 + (240\text{ mm})^2} = 388.89\text{ mm}$$

$$\overline{BC} = \sqrt{(240\text{ mm})^2 + (46.6\text{ mm})^2} = 244.48\text{ mm}$$

$$\underline{AB} = (306\text{ mm})\hat{i} + (-240\text{ mm})\hat{j}$$

$$\underline{F}_L = F_L \lambda_{CB}$$

$$\lambda_{CB} = \frac{\underline{CB}}{CB} = \frac{(240\text{ mm})\hat{i} + (-46.6\text{ mm})\hat{j}}{244.48\text{ mm}}$$

$$= (0.98)\hat{i} + (-0.191)\hat{j}$$

$$\therefore \underline{F}_L = F_L \lambda_{CB}$$

$$= 500\text{ N} \left[(0.98)\hat{i} + (-0.191)\hat{j} \right]$$

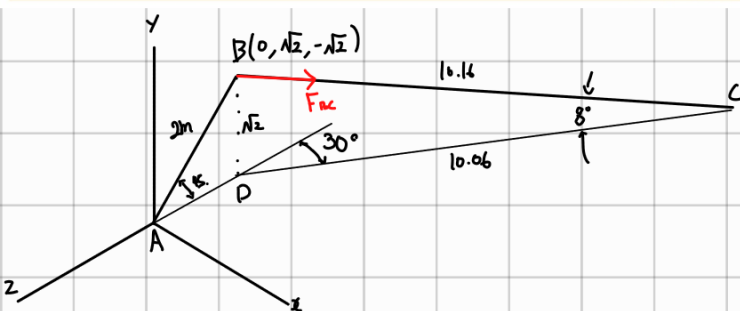
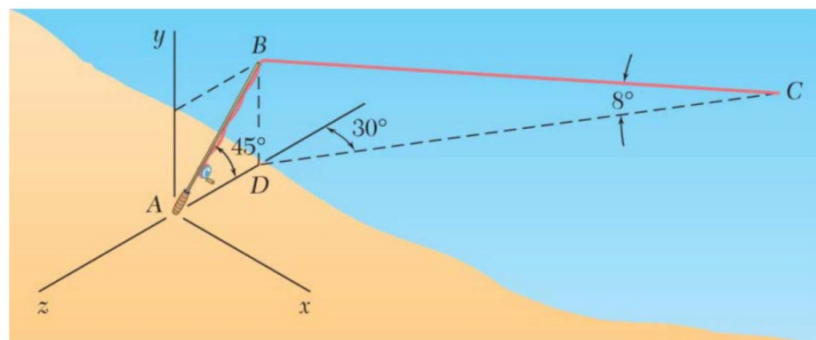
$$= (490\text{ N})\hat{i} + (-95.5\text{ N})\hat{j}$$

$$\therefore \underline{M}_A = \underline{AB} \times \underline{F}_L$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 306\text{ mm} & -240\text{ mm} & 0 \\ 490\text{ N} & -95.5\text{ N} & 0 \end{vmatrix}$$

$$\text{Ans.} = (112\text{ N}\cdot\text{m})\hat{k}$$

Problem 3.25: A 2-m-long fishing rod AB is securely anchored in the sand of a beach. After a fish takes the bait, the resulting force in the line is 30 N. Determine the moment about A of the force exerted by the line at B .



$$\tan 8^\circ = \frac{\sqrt{2}}{DC}$$

$$DC = \frac{\sqrt{2}}{\tan 8^\circ} = 10.06$$

$$\therefore \overline{BC} = \sqrt{(\sqrt{2})^2 + (10.06)^2} = 10.16$$

$$\therefore C(10.06 \sin 30^\circ, 0, -\sqrt{2} - 10.06 \cos 30^\circ)$$

$$C(5.03, 0, -10.13)$$

$$\overrightarrow{BC} = (5.03, -\sqrt{2}, -8.712) \quad 2029.0809 = 13.41$$

$$\overline{BC} = 10.16$$

$$\therefore \lambda_{BC} = \frac{\overrightarrow{BC}}{\overline{BC}} = (0.495, -0.139, -0.859)$$

$$\therefore \mathbf{F}_{BC} = F_{BC} \lambda_{BC} = (30 \text{ N})(0.495, -0.139, -0.859)$$

$$= (14.85 \text{ N}, -4.17 \text{ N}, -25.71 \text{ N})$$

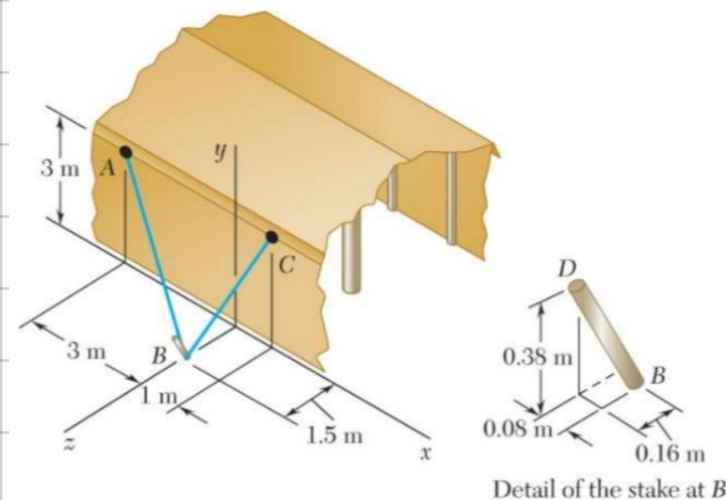
$$\overrightarrow{AB} = (0, \sqrt{2}, -\sqrt{2})$$

$$\mathbf{M}_A = \overrightarrow{AB} \times \mathbf{F}_{BC} = (0, \sqrt{2}, -\sqrt{2}) \times (14.85, -4.17, -25.71)$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & \sqrt{2} & -\sqrt{2} \\ 14.85 & -4.17 & -25.71 \end{vmatrix}$$

$$\text{Ans.} = (-42.2 \text{ N}\cdot\text{m})\hat{i} + (-21.0 \text{ N}\cdot\text{m})\hat{j} + (-21.0 \text{ N}\cdot\text{m})\hat{k}$$

Problem 3.41: Ropes AB and BC are two of the ropes used to support a tent. The two ropes are attached to a stake at B . If the tension in rope AB is 540 N, determine (a) the angle between rope AB and the stake, (b) the projection on the stake of the force exerted by rope AB at point B .



$$T_{AB} = 540 \text{ N}$$

$$\overrightarrow{BA} = (-3, 3, -1.5)$$

$$\overline{BA} = (3^2 + 3^2 + 1.5^2)^{0.5} = 4.5$$

$$\lambda_{BA} = \frac{\overrightarrow{BA}}{\overline{BA}} = \left(-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)$$

$$\overrightarrow{BD} = (-0.08, 0.38, 0.16)$$

$$\overline{BD} = (0.08^2 + 0.38^2 + 0.16^2)^{0.5} = 0.42$$

$$\lambda_{BD} = \frac{\overrightarrow{BD}}{\overline{BD}} = (0.190, 0.905, 0.381)$$

$$\lambda_{BA} \cdot \lambda_{BD} = \left(-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right) \cdot (0.190, 0.905, 0.381)$$

$$= 0.60 = |\lambda_{BA}| |\lambda_{BD}| \cos \theta$$

$$\therefore \cos \theta = 0.6$$

$$\text{a) } \theta = 53^\circ$$

$$T_{BA} = T_{AB} \lambda_{BA} = 540 \text{ N} \left(-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)$$

$$\therefore T_{BA} \cdot \lambda_{BD} = 540 \text{ N} \left(-\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right) \cdot (0.190, 0.905, 0.381)$$

$$\text{b) } = T_{AB} \times 1 \times \cos \theta = 540 \times \cos(53^\circ) = 324 \text{ N}$$

Problem 3.47: A crane is oriented so that the end of the 25-m boom AO lies in the yz plane. At the instant shown, the tension in cable AB is 4 kN. Determine the moment about each of the coordinate axes of the force exerted on A by cable AB .

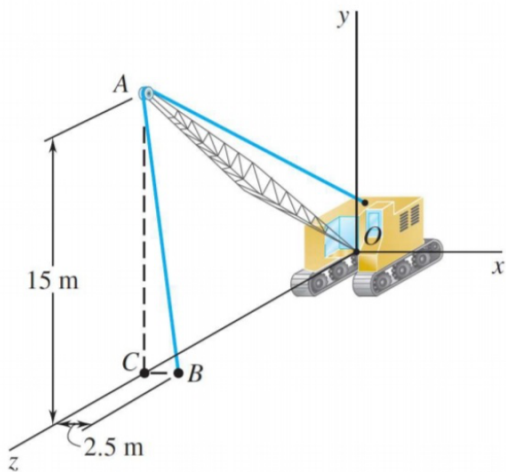
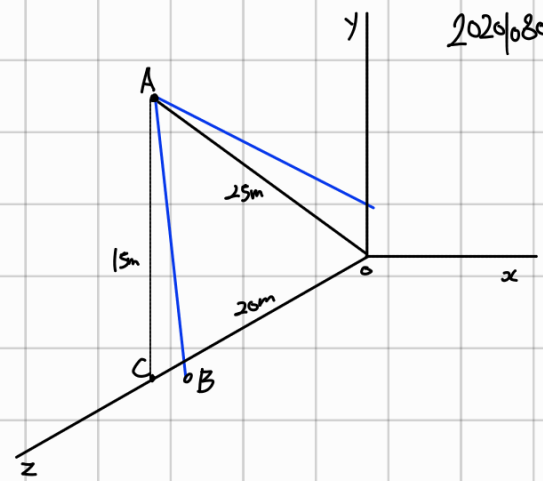


Fig. P3.47



$$\vec{OA} = (0, 15, 20)$$

$$OA = 25 \text{ m}$$

$$\lambda_{OA} = \left(0, \frac{3}{5}, \frac{4}{5}\right)$$

$$\vec{AB} = (2.5, -15, 0)$$

$$AB = (2.5^2 + 15^2)^{0.5} = 15.20 \text{ m}$$

$$\lambda_{AB} = \frac{\vec{AB}}{AB} = (0.1645, -0.9868, 0)$$

$$F = F \lambda_{AB} = 4 \text{ kN} (0.1645, -0.9868, 0)$$

$$\vec{OA} \times F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 15 & 20 \\ 0.6580 \text{ kN} & -3.947 \text{ kN} & 0 \end{vmatrix}$$

$$= (18.94 \text{ kN} \cdot \text{m}) \hat{i} + (13.16 \text{ kN} \cdot \text{m}) \hat{j} + (-9.87 \text{ kN} \cdot \text{m}) \hat{k}$$

$$\text{Ans.} \left(\begin{array}{l} \therefore M_x = 18.9 \text{ kN} \cdot \text{m} \\ M_y = 13.2 \text{ kN} \cdot \text{m} \\ M_z = -9.87 \text{ kN} \cdot \text{m} \end{array} \right)$$

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Problem 3.57: The frame ACD is hinged at A and D and is supported by a cable that passes through a ring at B and is attached to hooks at G and H . Knowing that the tension in the cable is 450 N , determine the moment about the diagonal AD of the force exerted on the frame by portion BH of the cable.

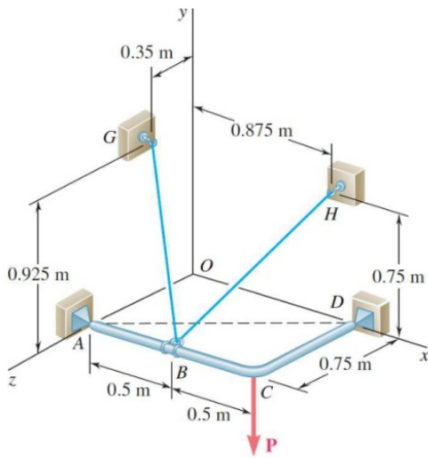


Fig. P3.57

$$\vec{AB} = 0.5\text{ m } \hat{i}$$

$$\vec{BH} = 0.375\text{ m } \hat{i} + 0.75\text{ m } \hat{j} - 0.75\text{ m } \hat{k}$$

$$|\vec{BH}| = (0.375^2 + 0.75^2 + 0.75^2)^{0.5} = \frac{9}{8}\text{ m}$$

$$\lambda_{BH} = \frac{\vec{BH}}{|\vec{BH}|} = \frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} - \frac{2}{3}\hat{k}$$

$$\vec{T}_{BH} = T_{BH} \lambda_{BH} = 450\text{ N} \left(\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} - \frac{2}{3}\hat{k} \right) = 150\text{ N } \hat{i} + 300\text{ N } \hat{j} - 300\text{ N } \hat{k}$$

$$\begin{aligned} \vec{M}_A &= \vec{AB} \times \vec{T}_{BH} = (0.5\text{ m } \hat{i}) \times (150\text{ N } \hat{i} + 300\text{ N } \hat{j} - 300\text{ N } \hat{k}) \\ &= 150\text{ N} \cdot \text{m } \hat{k} - 150\text{ N} \cdot \text{m } \hat{j} \\ &= -150\text{ N} \cdot \text{m } \hat{j} + 150\text{ N} \cdot \text{m } \hat{k} \end{aligned}$$

$$\vec{AD} = 1\text{ m } \hat{i} - 0.75\text{ m } \hat{k}$$

$$|\vec{AD}| = (1^2 + 0.75^2)^{0.5} = \frac{5}{4}\text{ m}$$

$$\lambda_{AD} = \frac{\vec{AD}}{|\vec{AD}|} = \frac{4}{5}\hat{i} - \frac{3}{5}\hat{k}$$

$$\therefore M_{AD} = \lambda_{AD} \cdot M_A = \left(\frac{4}{5}\hat{i} - \frac{3}{5}\hat{k} \right) \cdot (-150\text{ N} \cdot \text{m } \hat{j} + 150\text{ N} \cdot \text{m } \hat{k})$$

$$\text{ans.} = -90\text{ N} \cdot \text{m}$$

Problem 3.74: A piece of plywood in which several holes are being drilled successively has been secured to a workbench by means of two nails. Knowing that the drill exerts a $12\text{-N}\cdot\text{m}$ couple on the piece of plywood, determine the magnitude of the resulting forces applied to the nails if they are located (a) at A and B , (b) at B and C , (c) at A and C .

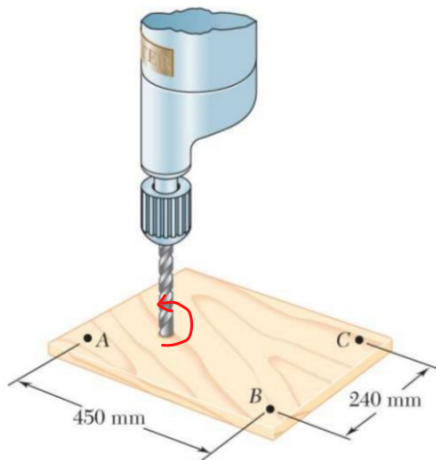
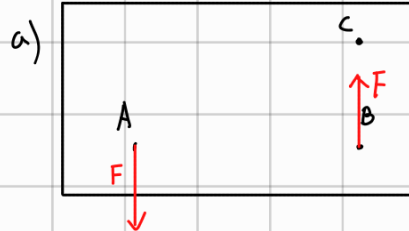
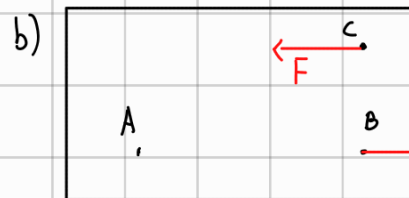


Fig. P3.74



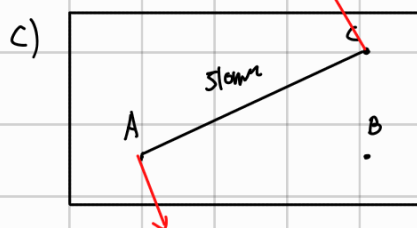
$$Fd = 12\text{ N} \cdot \text{m}$$

$$\therefore F = \frac{12\text{ N} \cdot \text{m}}{d} = \frac{12\text{ N} \cdot \text{m}}{0.45\text{ m}} = \frac{80}{3}\text{ N}$$



$$Fd = 12\text{ N} \cdot \text{m}$$

$$F = \frac{12\text{ N} \cdot \text{m}}{d} = \frac{12\text{ N} \cdot \text{m}}{0.24\text{ m}} = 50\text{ N}$$



$$Fd = 12\text{ N} \cdot \text{m}$$

$$F = \frac{12\text{ N} \cdot \text{m}}{d} = \frac{12\text{ N} \cdot \text{m}}{0.51\text{ m}} = \frac{400}{17}\text{ N}$$

$$AC = (450^2 + 240^2)^{0.5} = 510\text{ mm}$$

Problem 3.75: The shafts of an angle drive are acted upon by the two couples shown. Replace the two couples with a single equivalent couple, specifying its magnitude and the direction of its axis.

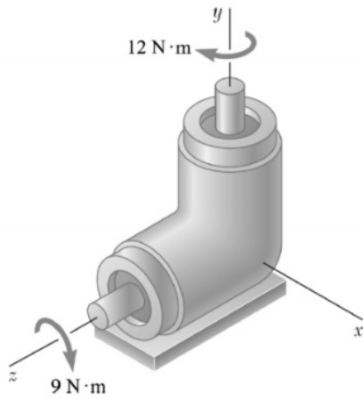
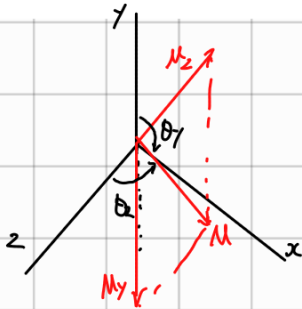


Fig. P3.75



$$M_y = -12 \text{ N} \cdot \text{m} \hat{j}$$

$$M_z = -9 \text{ N} \cdot \text{m} \hat{k}$$

$$\therefore \underline{M} = M_y + M_z$$

$$= -12 \text{ N} \cdot \text{m} \hat{j} - 9 \text{ N} \cdot \text{m} \hat{k} \quad y-z \text{ plane}$$

$$|M| = (12^2 + 9^2)^{0.5} = 15 \text{ N} \cdot \text{m}$$

$$M \cdot \hat{j} = (-12 \text{ N} \cdot \text{m} \hat{j} - 9 \text{ N} \cdot \text{m} \hat{k}) \cdot \hat{j} = 15 \times 1 \times \cos \theta_y$$

$$\cos \theta_y = -\frac{12}{15} = -\frac{4}{5}$$

$$\therefore \theta_y = 143.1^\circ$$

$$M \cdot \hat{k} = (-12 \text{ N} \cdot \text{m} \hat{j} - 9 \text{ N} \cdot \text{m} \hat{k}) \cdot \hat{k} = 15 \times 1 \times \cos \theta_z$$

$$\cos \theta_z = -\frac{9}{15} = -\frac{3}{5}$$

$$\therefore \theta_z = 126.8^\circ$$

$$M \cdot \hat{i} = 0$$

$$\therefore \theta_x = 90^\circ$$

Problem 3.99: An antenna is guyed by three cables as shown. Knowing that the tension in cable AB is 1.44 kN, replace the force exerted at A by cable AB with an equivalent force-couple system at the center O of the base of the antenna.

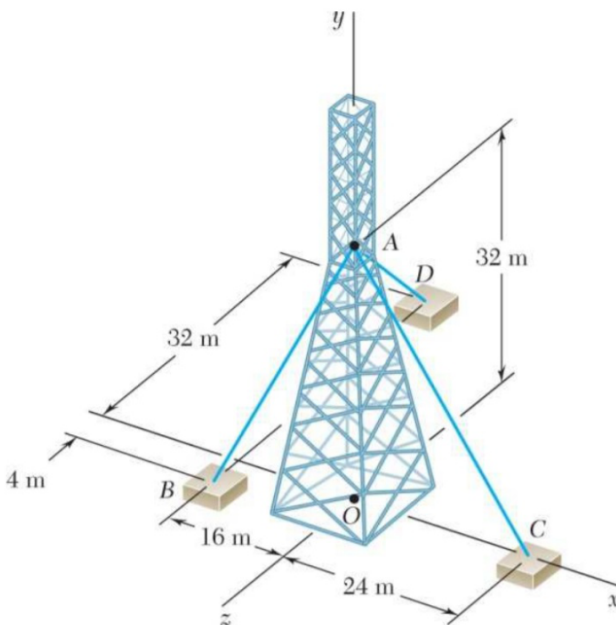


Fig. P3.99

$$\vec{AB} = -16\text{m} \hat{i} - 32\text{m} \hat{j} + 4\text{m} \hat{k}$$

$$|\vec{AB}| = (16^2 + 32^2 + 4^2)^{0.5} = 36\text{m}$$

$$\lambda_{AB} = \frac{\vec{AB}}{|\vec{AB}|} = -\frac{4}{9} \hat{i} - \frac{8}{9} \hat{j} + \frac{1}{9} \hat{k}$$

$$\therefore \vec{T}_{AB} = T_{AB} \lambda_{AB}$$

$$= 1.44 \text{ kN} \left(-\frac{4}{9} \hat{i} - \frac{8}{9} \hat{j} + \frac{1}{9} \hat{k} \right)$$

$$\text{Ans} = -\frac{16}{25} \text{ kN} \hat{i} - \frac{32}{25} \text{ kN} \hat{j} + \frac{4}{25} \text{ kN} \hat{k}$$

$$\vec{OA} = 32\text{m} \hat{j}$$

$$\underline{M}_O = \vec{OA} \times \vec{T}_{AB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 32\text{m} & 0 \\ -\frac{16}{25} \text{ kN} & -\frac{32}{25} \text{ kN} & \frac{4}{25} \text{ kN} \end{vmatrix}$$

$$\text{Ans} = 5.12 \text{ kN} \cdot \text{m} \hat{i} + 20.48 \text{ kN} \cdot \text{m} \hat{k}$$

Problem 3.101: A 3-m-long beam is subjected to a variety of loadings. (a) Replace each loading with an equivalent force-couple system at end A of the beam. (b) Which of the loadings are equivalent?

Problem 3.103: Determine the single equivalent force and the distance from point A to its line of action for the beam and loading of (a) Prob. 3.101a, (b) Prob. 3.101b, (c) Prob. 3.101h.

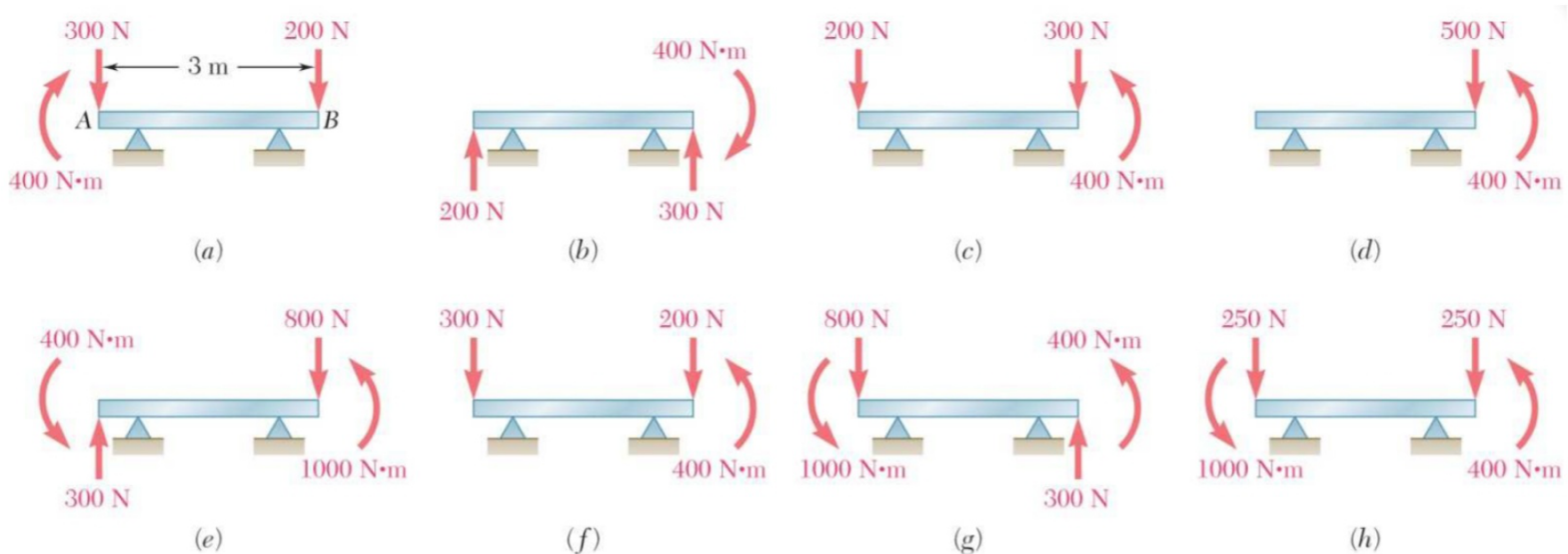
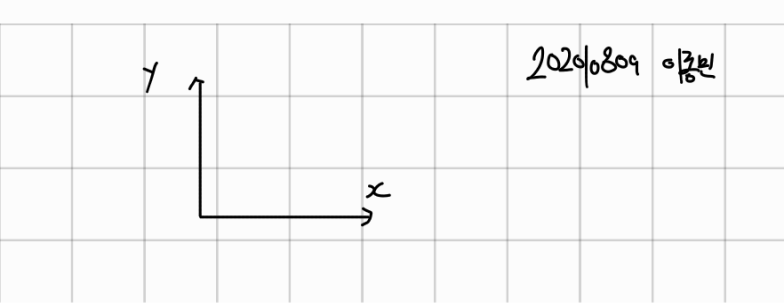
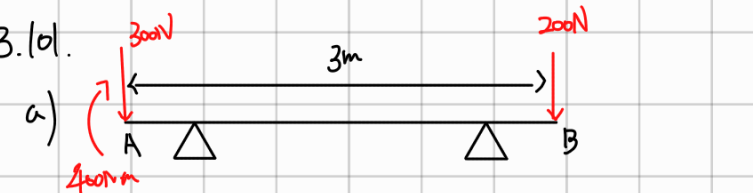
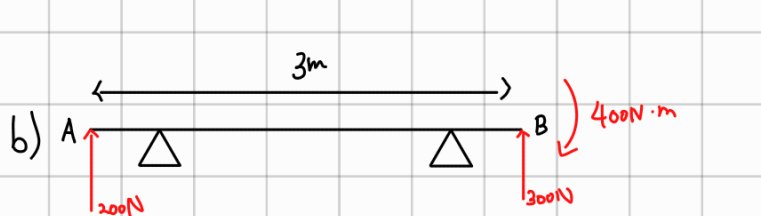


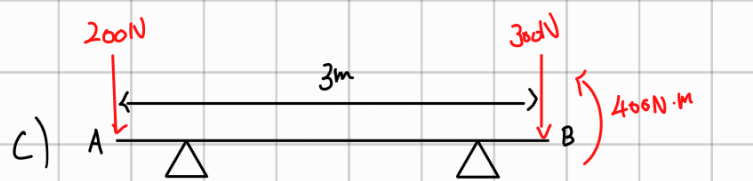
Fig. P3.101/103



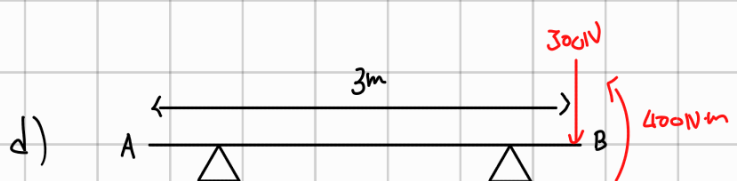
a) $P_A = 300\text{ N} - 200\text{ N} = 100\text{ N}$
 $M_A = -400\text{ N}\cdot\text{m} - 200\text{ N} \times 3\text{ m} = -1000\text{ N}\cdot\text{m}$



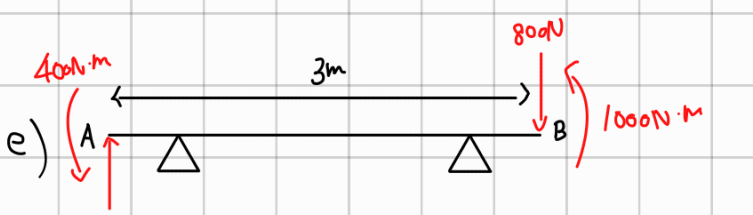
a) $P_A = 200\text{ N} + 300\text{ N} = 500\text{ N}$
 $M_A = -400\text{ N}\cdot\text{m} + 300\text{ N} \times 3\text{ m} = 500\text{ N}\cdot\text{m}$



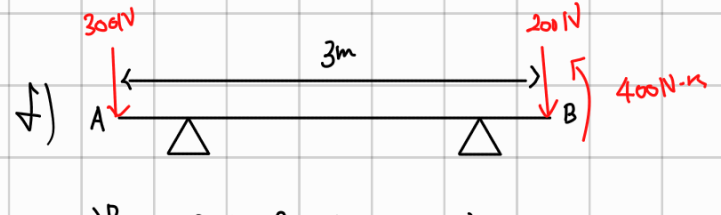
a) $P_A = 200\text{ N} - 300\text{ N} = -100\text{ N}$
 $M_A = 400\text{ N}\cdot\text{m} - 300\text{ N} \times 3\text{ m} = -500\text{ N}\cdot\text{m}$



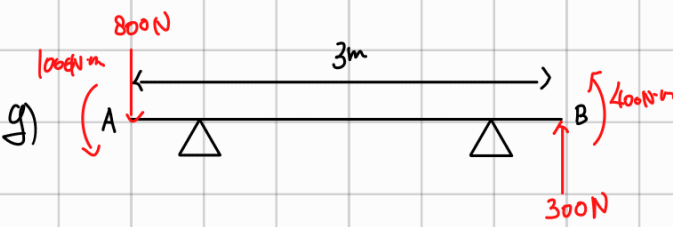
$P_A = -300\text{ N}$
 $M_A = 400\text{ N}\cdot\text{m} - 300\text{ N} \times 3\text{ m} = -500\text{ N}\cdot\text{m}$



a) $P_A = 300\text{ N} - 800\text{ N} = -500\text{ N}$
 $M_A = 400\text{ N}\cdot\text{m} + 1000\text{ N}\cdot\text{m} - 800\text{ N} \times 3\text{ m} = -200\text{ N}\cdot\text{m}$



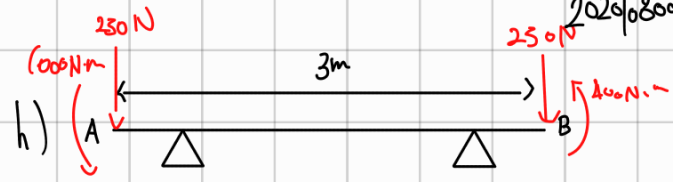
a) $P_A = 300\text{ N} + 200\text{ N} = 500\text{ N}$
 $M_A = 400\text{ N}\cdot\text{m} - 200\text{ N} \times 3\text{ m} = -200\text{ N}\cdot\text{m}$



a)

$$P_A = -800\text{N} + 300\text{N} = -500\text{N}$$

$$M_A = 1000\text{N}\cdot\text{m} + 400\text{N}\cdot\text{m} + 300\text{N} \cdot 3\text{m} = 2300\text{N}\cdot\text{m}$$



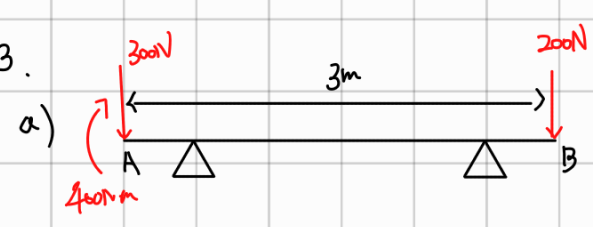
a)

$$P_A = -250\text{N} - 250\text{N} = -500\text{N}$$

$$M_A = 1000\text{N}\cdot\text{m} - 400\text{N}\cdot\text{m} - 250\text{N} \cdot 3\text{m} = -650\text{N}\cdot\text{m}$$

b) a), e) 의 경우와 같
 $P_A = -500\text{N}$
 $M_A = -1000\text{N}\cdot\text{m}$ 3월 02일

3.103.



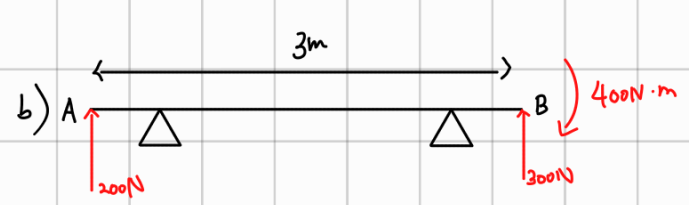
$$P_A = 300\text{N} - 200\text{N} = -500\text{N}$$

$$M_A = -400\text{N}\cdot\text{m} - 200\text{N} \cdot 3\text{m} = -1000\text{N}\cdot\text{m}$$

$$P = -500\text{N}$$

$$M_A = P \cdot x = -1000\text{N}\cdot\text{m}$$

$$x = \frac{-1000\text{N}\cdot\text{m}}{P} = \frac{-1000\text{N}\cdot\text{m}}{-500\text{N}} = 2\text{m}$$



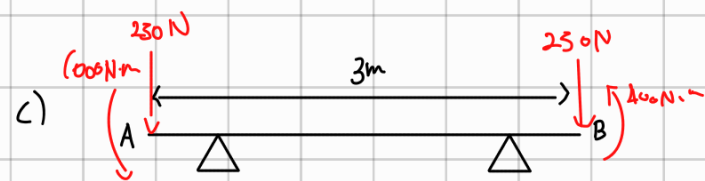
$$P_A = 200\text{N} + 300\text{N} = 500\text{N}$$

$$M_A = -400\text{N}\cdot\text{m} + 300\text{N} \cdot 3\text{m} = 500\text{N}\cdot\text{m}$$

$$P = 500\text{N}$$

$$M_A = P \cdot x = 500\text{N}\cdot\text{m}$$

$$x = \frac{500\text{N}\cdot\text{m}}{P} = \frac{500\text{N}\cdot\text{m}}{500\text{N}} = 1\text{m}$$



$$P_A = -250\text{N} - 250\text{N} = -500\text{N}$$

$$M_A = 1000\text{N}\cdot\text{m} - 400\text{N}\cdot\text{m} - 250\text{N} \cdot 3\text{m} = -650\text{N}\cdot\text{m}$$

$$P = -500\text{N}$$

$$M_A = P \cdot x = -650\text{N}\cdot\text{m}$$

$$x = \frac{-650\text{N}\cdot\text{m}}{P} = \frac{-650\text{N}\cdot\text{m}}{-500\text{N}} = 1.3\text{m}$$

Problem 3.120: Two 150-mm-diameter pulleys are mounted on line shaft AD . The belts at B and C lie in vertical planes parallel to the yz -plane. Replace the belt forces shown with an equivalent force-couple system at A .

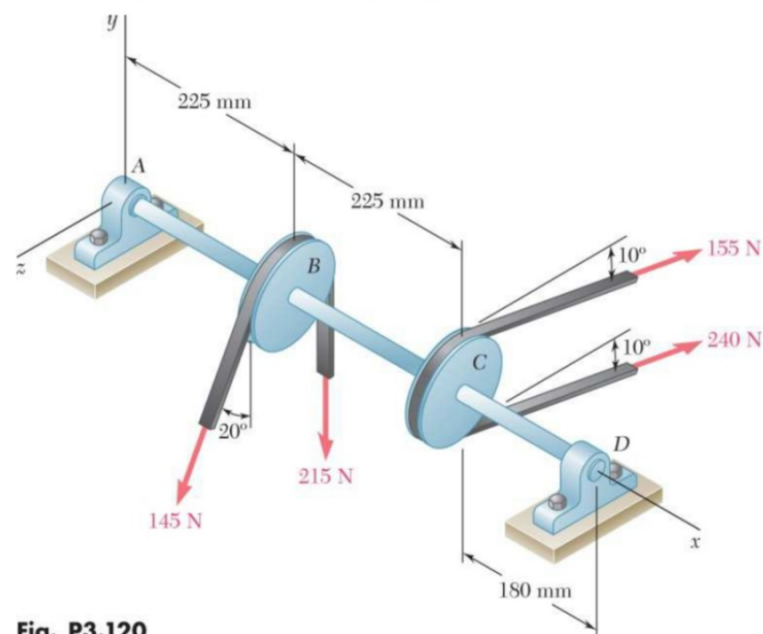


Fig. P3.120

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$$\begin{aligned} \underline{F}_B &= 145\text{N}(-\cos 20^\circ \hat{j} + \sin 20^\circ \hat{k}) - 215\text{N}(-\hat{j}) \\ &= -351.26\text{N} \hat{j} + 49.592\text{N} \hat{k} \\ \underline{M}_B &= 145\text{N} \times 75\text{mm} - 215\text{N} \times 75\text{mm} \\ &= -5.25\text{N} \cdot \text{m} \hat{i} \end{aligned}$$

$$\begin{aligned} \underline{F}_C &= (155\text{N} + 240\text{N})(-\sin 10^\circ \hat{j} - \cos 10^\circ \hat{k}) \\ &= -68.591\text{N} \hat{j} - 389.00\text{N} \hat{k} \\ \underline{M}_C &= (-155\text{N} \times 75\text{mm} + 240\text{N} \times 75\text{mm}) \hat{i} \\ &= 6.375\text{N} \cdot \text{m} \hat{i} \end{aligned}$$

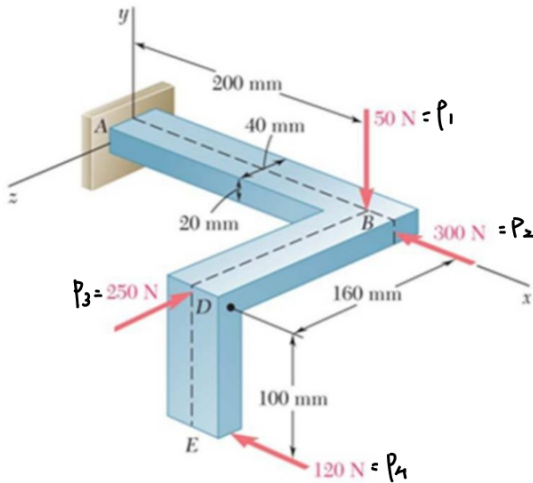
$$\therefore \underline{F}_A = \underline{F}_B + \underline{F}_C = -419.85\text{N} \hat{j} - 339.41\text{N} \hat{k}$$

$$\begin{aligned} \underline{M}_A &= \underline{M}_B + \underline{M}_C + \underline{AB} \times \underline{F}_B + \underline{AC} \times \underline{F}_C \\ &= \underline{M}_B + \underline{M}_C + (0.225\text{m} \hat{i}) \times (-351.26\text{N} \hat{j} + 49.592\text{N} \hat{k}) \\ &\quad + (0.45\text{m} \hat{i}) \times (-68.591\text{N} \hat{j} - 389.00\text{N} \hat{k}) \\ &= \underline{M}_B + \underline{M}_C + (-79.034\text{N} \cdot \text{m} \hat{k} - 11.158\text{N} \cdot \text{m} \hat{j}) \\ &\quad + (-30.866\text{N} \cdot \text{m} \hat{k} + 175.05\text{N} \cdot \text{m} \hat{j}) \end{aligned}$$

$$\text{ans. } \underline{M}_A = (1.12\text{N} \cdot \text{m}) \hat{i} + (163.9\text{N} \cdot \text{m}) \hat{j} + (-109.9\text{N} \cdot \text{m}) \hat{k}$$

$$\underline{P}_A = \underline{F}_A = \underline{F}_B + \underline{F}_C = -419.85\text{N} \hat{j} - 339.41\text{N} \hat{k}$$

Problem 3.124: Four forces are applied to the machine component ABDE as shown. Replace these forces with an equivalent force-couple system at A.



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$$\begin{aligned} \mathbf{P}_A &= \mathbf{P}_1 + \mathbf{P}_2 + \mathbf{P}_3 + \mathbf{P}_4 \\ &= (-50\text{ N})\hat{j} + (-300\text{ N})\hat{i} + (-250\text{ N})\hat{i} + (-120\text{ N})\hat{j} \\ &= (-420\text{ N})\hat{i} + (-50\text{ N})\hat{j} + (-250\text{ N})\hat{k} \end{aligned}$$

$$\mathbf{r}_1 = (0.2\text{ m})\hat{i}$$

$$\mathbf{r}_2 = (0.22\text{ m})\hat{i}$$

$$\mathbf{r}_3 = (0.2\text{ m})\hat{i} + (0.16\text{ m})\hat{k}$$

$$\mathbf{r}_4 = (0.2\text{ m})\hat{i} + (-0.1\text{ m})\hat{j} + (0.16\text{ m})\hat{k}$$

$$\mathbf{M}_A = \mathbf{r}_1 \times \mathbf{P}_1 + \mathbf{r}_2 \times \mathbf{P}_2 + \mathbf{r}_3 \times \mathbf{P}_3 + \mathbf{r}_4 \times \mathbf{P}_4$$

$$= (0.2\text{ m})\hat{i} \times (-50\text{ N})\hat{j} + (0.22\text{ m})\hat{i} \times (-300\text{ N})\hat{i}$$

$$+ \{(0.2\text{ m})\hat{i} + (0.16\text{ m})\hat{k}\} \times (-250\text{ N})\hat{k}$$

$$+ \{(0.2\text{ m})\hat{i} + (-0.1\text{ m})\hat{j} + (0.16\text{ m})\hat{k}\} \times (-120\text{ N})\hat{j}$$

$$= (-10\text{ N}\cdot\text{m})\hat{k} + 0 + (50\text{ N}\cdot\text{m})\hat{j} + (-12\text{ N}\cdot\text{m})\hat{k} + (-19.2\text{ N}\cdot\text{m})\hat{j}$$

WNS.

$$\mathbf{M}_A = (30.8\text{ N}\cdot\text{m})\hat{j} + (-22\text{ N}\cdot\text{m})\hat{k}$$

$$\mathbf{P}_A = (-420\text{ N})\hat{i} + (-50\text{ N})\hat{j} + (-250\text{ N})\hat{k}$$